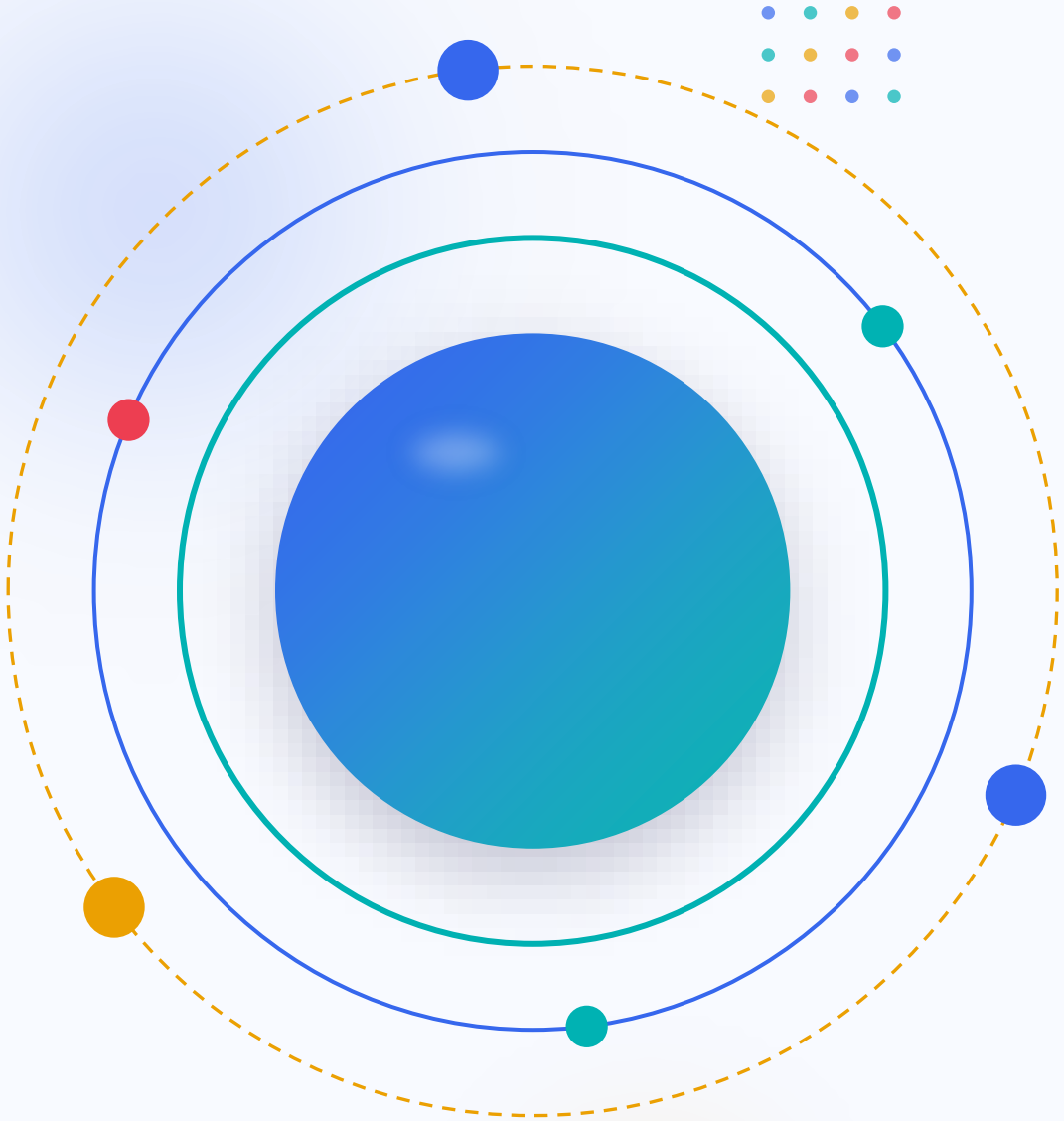
ROUND 1 · PROBLEM 3

NovaSense: predict failures before they happen

An AI co-pilot that watches every machine and tells the team exactly what to fix next.

Ctrl.Alt.Elite

Umer Basith · Muizzur Rahman



Machines fail without warning, and teams find out only after production stops.

Every hour of downtime costs lakhs in lost output.

20%

of productive time lost to unplanned downtime

03 · OUR SOLUTION

One platform, three superpowers

NovaSense turns raw sensor readings into early warnings and clear actions.



01

Predict

A gradient-boosting model scores failure risk from temperature, vibration, load and run hours.



02

Alert

High-risk machines raise instant, prioritised alerts with the main cause.



03

Explain

An AI assistant answers questions in plain English using live data.

From sensor to decision in under a second

1

Sensors / CSV

Readings arrive live or as an upload

2

FastAPI backend

Validates input and stores history in SQLite

3

ML model

Scores risk and explains the top factors

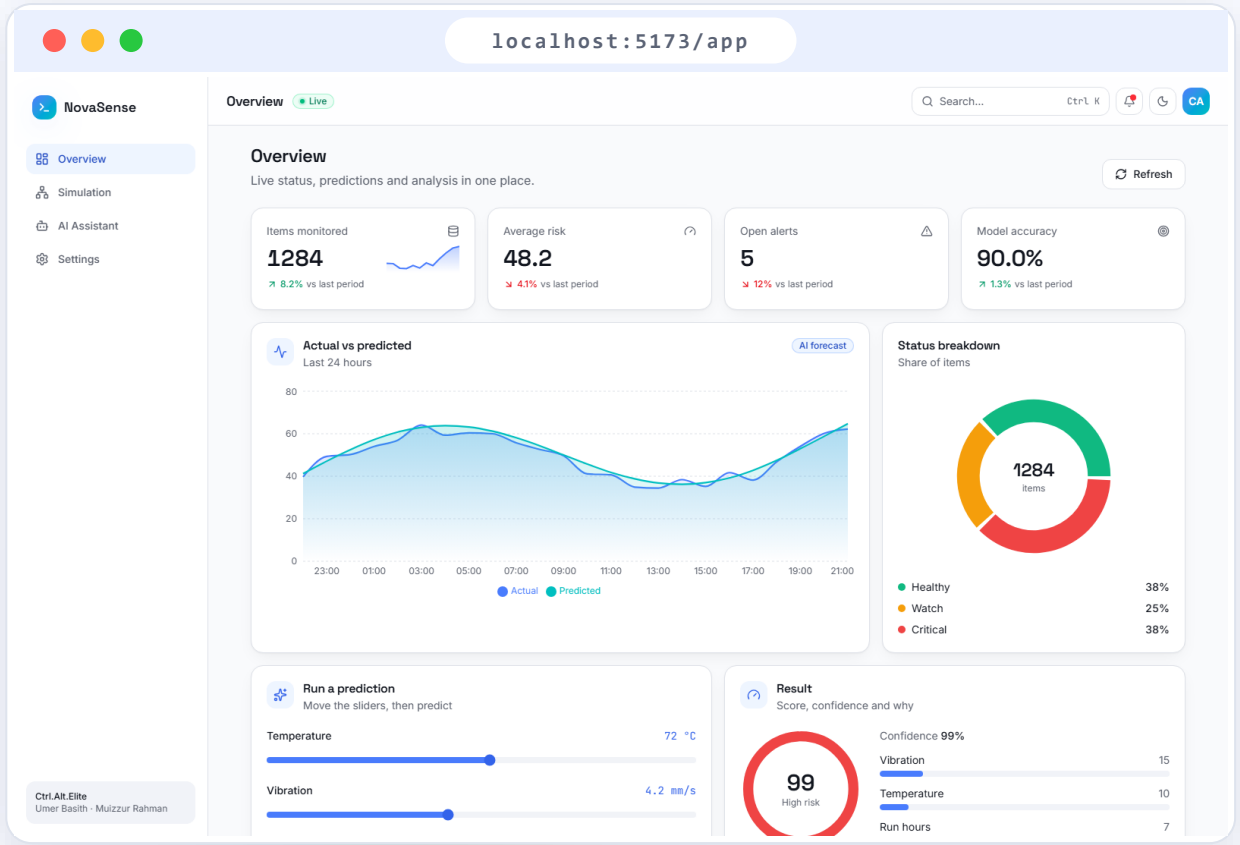
4

React dashboard

Charts, alerts and the AI assistant

05 · LIVE DEMO

The dashboard the team actually uses



localhost:5173/app — running live

1. Live KPIs and 24-hour forecast
2. Risk score with reasons
3. Status mix at a glance
4. Upload any CSV to analyse

Measured, not guessed

90%

Accuracy on 1,000 unseen rows

0.96

ROC-AUC

<1s

Time to score a machine

6/6

Automated tests passing

Model trained on a synthetic dataset modelled on real sensor ranges. Real plant data would be the next step.

Before vs after NovaSense

Today

- ✕ Fix after it breaks
- ✕ Paper logs
- ✕ Experts guess the cause
- ✕ Hours of downtime

With NovaSense

- ✓ Warned hours earlier
- ✓ Live dashboard
- ✓ Top causes shown
- ✓ Planned, short stops

How we compare

★ BEST

Feature	Manual checks	Basic alarms	NovaSense
Predicts early	No	No	Yes
Explains cause	Sometimes	No	Yes
Plain-English assistant	No	No	Yes
Cost to start	High	Medium	Low

What comes next



Today

Working prototype with model, dashboard and assistant

Month 1

Pilot on one real production line

Month 3

Edge device + live sensor stream

Month 6

Multi-plant rollout and mobile alerts

Thank you

NovaSense: from raw signals to the right action, in seconds.

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Questions? We'd love to show you the live demo.